

EC300:Control Systems

Project-Title: ADCS-1 - Single-Axis CubeSat Attitude Control Simulator

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Abstract:

This project designs and builds a single-axis attitude control system for a simulated CubeSat, using a reaction wheel to demonstrate torque-based pointing control without any propulsion or aerodynamic surfaces. The system plant is modeled as a double integrator, derived from conservation of angular momentum between a rotating body and an internal flywheel.

The project covers full analytical controller design: proportional-derivative and lead compensator design targeting a specified phase margin, verified through Bode plot, root locus, and Nyquist stability analysis. A Python-based simulation validates controller performance before hardware implementation, including step response characterization and rejection of a constant disturbance torque.

Additional work includes Kalman filter-based state estimation for improved angle tracking, empirical characterization and feedforward compensation of bearing friction, and a simulated demonstration of reaction wheel saturation and desaturation behavior.